

FUNCTIONAL SPECIALIZATION ACROSS MOUSE AUDITORY
CORTICAL SUBREGIONS

by

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A THESIS

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The auditory cortex (AC) is organized into distinct subregions thought to process different aspects of sound, but the specific functional roles of secondary areas remain unclear. This project studies how four subregions of the mouse AC represent sounds of increasing acoustic complexity using activity responses from groups of neurons. Population responses were recorded simultaneously from the primary (AudP), dorsal (AudD), ventral (AudV), and posterior (AudPo) subregions while mice passively listened to pure tones (PT), amplitude-modulated (AM) white noise, environmental natural sounds, and speech syllables. Using analyses of how populations of neurons organize sound information and how reliably those responses can be distinguished, the results showed that AC subregions differ in how they represent sound. AudP showed broad and consistent engagement across all sound types, serving as a reliable baseline for interpreting the secondary subregions. AudV showed strong representations of simple sounds but weaker performance for complex and categorical sounds. AudPo showed the strongest discrimination of natural sounds and speech, consistent with a preferential role in higher-order acoustic processing. AudD showed consistently limited performance across all stimulus types, though this may reflect a mismatch with the stimulus set rather than a general processing deficit. These results are most consistent with a distributed processing framework, in which subregions share the labor of sound processing with contributions shifting depending on acoustic content. Yet, the stronger performance of AudV for simple sounds and AudPo for complex sounds offers partial support for a hierarchical organization as well. A clearer understanding of how the AC is organized has broader implications for research on hearing loss, auditory processing disorders, and the development of auditory prosthetics.

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1. Introduction

The brain's ability to make sense of sound, from a simple tone to a complex sentence, depends on the auditory cortex (AC). The AC is divided into distinct subregions, including a primary and several secondary areas thought to process different aspects of sound. The primary AC (AudP) is the best-characterized subregion, known to be organized by sound frequency, with neurons tuned to specific frequencies arranged systematically across the cortical surface (Joachimsthaler et al., 2014). The dorsal (AudD), ventral (AudV), and posterior (AudPo) auditory areas make up the secondary AC subregions, but their functional roles remain much less understood (Hackett, 2011). What remains unclear is how these subregions divide the labor of sound processing, and whether that division shifts as sounds become more complex. Three frameworks have been proposed in the literature, each supported by empirical evidence and making distinct predictions about the functional roles of each subregion.

A parallel processing organization proposes that secondary AC subregions are each exclusively specialized for a distinct **acoustic content**¹, processing it independently and in parallel with other subregions. Evidence from primate AC supports two such pathways, one selective for the location of a sound in space and one selective for sound identity, each targeting distinct regions of the brain (Rauschecker and Tian, 2000). Under this model, each secondary subregion holds a fixed specialization that does not change with the type or complexity of the sound being heard, with little overlap in what each area contributes across stimulus types.

A distributed processing model proposes that sound processing is shared across subregions, with each contributing partial and overlapping specializations rather than exclusive fixed roles (Hackett, 2011; Winer and Lee, 2007). Under this framework, subregions are densely

¹ Bolded terms are defined in the Glossary (Section 8)

interconnected and each contributes distinct information, meaning no single area is fully redundant with another, nor solely responsible for processing a given sound (Hackett, 2011; Winer and Lee, 2007). This model predicts that no subregion should dominate uniformly across all sound types, and that the pattern of which area contributes most should change as sounds become more complex.

A hierarchical model proposes that sound processing is organized as a sequence of stages, with AudP receiving the earliest input and passing information to secondary subregions, which connect further to higher-level brain areas (Hackett, 2011). In mice, AudP neurons respond most strongly to simple sounds, while neurons in secondary subregions show greater sensitivity to more complex sounds (Joachimsthaler et al., 2014). Speech-selective neurons in particular are more prevalent in the dorso-posterior region than in AudP (Mohn et al., 2024). Under this framework, AudD, AudV, and AudPo would respond more selectively than AudP to complex natural sounds and speech, and less reliably to simple sounds.

These three frameworks are not mutually exclusive, and each has received empirical support (Hackett, 2011). Distinguishing between them requires methods that can characterize how entire populations of neurons represent sound, rather than one neuron at a time. Single-neuron studies have characterized individual response properties (Joachimsthaler et al., 2014), but the collective activity of a population carries structure that cannot be seen at the single neuron level. The development of **Neuropixels** probes, a type of high-density recording electrode, made it feasible to record from hundreds of neurons across multiple cortical subregions simultaneously in awake animals (Jun et al., 2017), generating the kind of large-scale population data needed to distinguish between these frameworks.

The present study analyzes Neuropixels recordings from head-fixed mice passively listening to four categories of sounds, ranging from simple pure tones (PT) and amplitude-modulated (AM) white noise to complex environmental natural sounds and synthetic speech syllables, across four AC subregions: AudP, AudD, AudV, and AudPo. All experiments and data collection were performed by the Jaramillo lab at the University of Oregon. This study analyzes these recordings using two complementary approaches to study how each subregion represents sound. Encoding analyses ask how sound information is organized within a population, using dimensionality reduction to visualize the **neural manifolds** of **population responses** and quantify their **dimensionality**. Decoding analyses ask how accurately sound identity can be determined from its population responses, using regression to predict continuous stimulus values and classification to quantify pairwise stimulus **discriminability**. The results show functional specializations across subregions that shift with sound complexity, most consistent with a distributed processing framework while offering partial support for a hierarchical organization, with implications for how AC dysfunction is understood and treated.

2. Data and Experimental Design

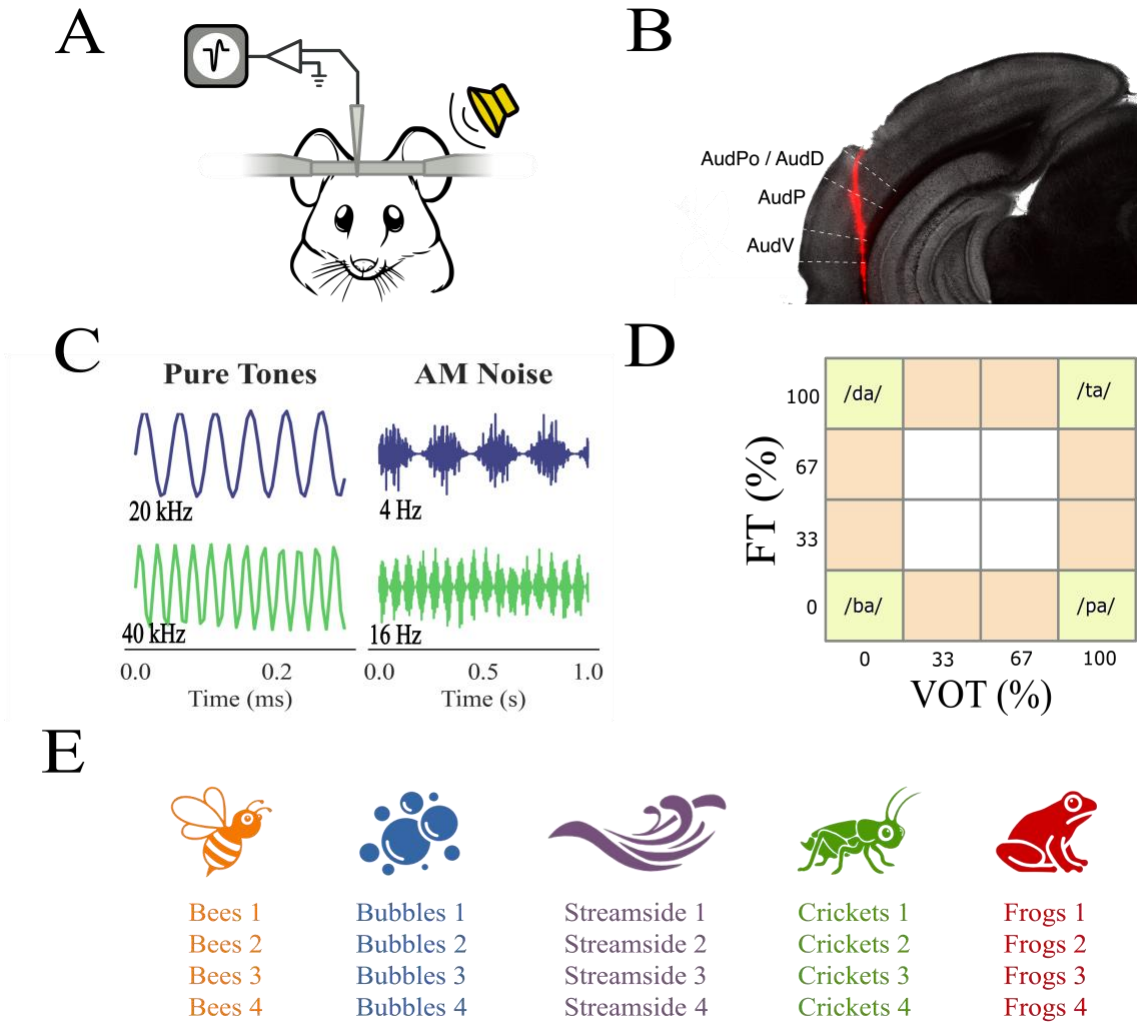


Figure 1: Experimental Design and Stimulus Set

Experimental design and auditory stimulus set. (A) Schematic of the head-fixed passive listening paradigm. Mice were head-fixed on a wheel and presented with sounds during neural recording. (B) Histological brain section showing an example Neuropixels probe track targeting AudP, AudV, AudD and AudPo, which is not directly visible at this angle. (C) Schematic waveforms for PT and AM stimulus classes. (D) Speech syllable stimulus space defined by voice onset time (VOT, x-axis) and formant transition (FT, y-axis), each ranging from 0 to 100% of maximum. Twelve syllables at boundary and corner positions were presented; canonical endpoints (/ba/, /da/, /pa/, /ta/) are labeled. (E) Natural sound stimulus set drawn from five environmental categories (frogs, crickets, streamside, bubbling, bees), with four exemplars per category.

This study combines two datasets of neural recordings from head-fixed mice listening to four sound types of increasing acoustic complexity. All experiments and data collection described in this section were performed by members of the Jaramillo Lab.

2.1 Stimulus Classes

2.1.1 Pure Tones (PT)

PT stimuli consisted of 16 frequencies logarithmically spaced between 2 and 40 kHz (Fig. 1C, left), presented for 100 ms at 60 and 70dB SPL. PT are the simplest stimulus class tested, isolating single spectral features.

2.1.2 Amplitude-Modulated (AM) White Noise

AM white noise stimuli consisted of 11 modulation rates logarithmically spaced between 4 and 128 Hz (Fig. 1C, right), presented for 500 ms at 60 dB SPL. Modulation rate is the frequency at which a sound's amplitude fluctuates per second. AM stimuli vary in temporal structure, representing an intermediate level of acoustic complexity.

2.1.3 Natural Sounds

Natural sound stimuli consisted of 20 four-second recordings drawn from five environmental categories (frogs, crickets, streamside, bubbling, and bees), with four non-overlapping exemplars per category (Fig. 1E). Unlike PT or AM noise, natural sounds vary in both spectral features and temporal structure, making them a more complex stimulus class.

2.1.4 Speech Syllables

Synthetic speech syllables were generated along two acoustic dimensions: voice onset time (VOT) and formant transition (FT) (Mohn et al., 2024). VOT is the delay between the

release of a stop consonant and the onset of vocal cord vibration, distinguishing voiced from unvoiced stops (e.g., /ba/ vs. /pa/). FT is the trajectory of the second formant frequency at syllable onset, distinguishing place of articulation (e.g., /ba/ vs. /da/). Stimuli were arranged along a grid defined by VOT and FT values of 0, 33, 67, and 100 percent of maximum, resulting in 12 unique syllables, 500 ms each (Fig. 1D). AudD was excluded from speech analyses due to insufficient neuron count for reliable **subsampling**.

2.2 Behavioral Paradigm

Mice were head-fixed on a wheel (Fig. 1A) for the duration of each recording session. Stimulus classes were presented in order of increasing complexity. Within each class, individual stimuli were presented in a randomized order, cycling through all stimuli once before repeating to prevent neural adaptation. PT, AM, and speech stimuli were each presented 20 times per session and natural sound stimuli were presented 10 times each.

2.3 Neural Recordings and Spike Sorting

Extracellular recordings were performed using **Neuropixels** 1.0 probes inserted acutely to target the entire primary and secondary AC simultaneously. Neural data were assigned to subregions histologically based on electrode track location aligned with the Allen Mouse Brain Atlas (Lein et al., 2007). Sound stimuli were time-locked to neural recordings for **firing rate** analysis. Raw voltage signals were processed using Kilosort for automated spike detection and clustering (Pachitariu et al., 2024), then manually curated using Phy (Rossant et al., 2016), with units accepted or rejected based on waveform shape, inter-spike interval distributions, and cluster separation.

3. Methods

3.1 Firing Rate Calculation and Normalization

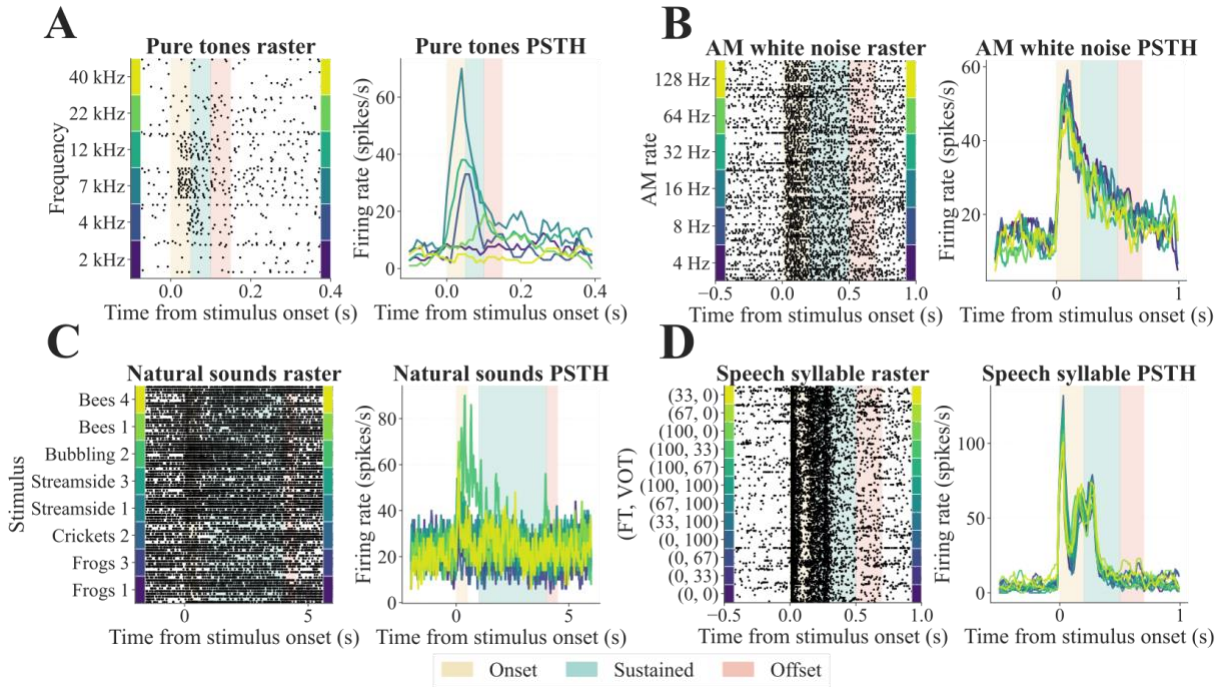


Figure 2: Single-Cell Neural Responses Across Stimulus Types and Analysis Windows

Example single-unit raster plots (left) and peri-stimulus time histograms (right, 10-ms boxcar smoothing) for representative cells responding to (A) pure tones, (B) AM white noise, (C) natural sounds, and (D) speech syllables. Trials are color-coded by stimulus identity. Shaded regions mark the three non-overlapping analysis windows: onset (yellow), sustained (teal), and offset (orange). Window boundaries differ by stimulus type to reflect each sound's temporal dynamics (see Section 3.1).

Firing rates were calculated from spike times using three non-overlapping analysis windows, onset, sustained, and offset. PT windows were onset [0–50 ms], sustained [50–100 ms], and offset [100–150 ms]. AM and speech windows were onset [0–200 ms], sustained [200–500 ms], and offset [500–700 ms]. Natural sounds windows were onset [0–500 ms], sustained [1000–4000 ms], and offset [4000–4500 ms]. These **spike windows** were chosen to reflect each

sound's characteristic temporal dynamics (Fig. 2). For each neuron, spike counts within each window were divided by window duration to compute mean firing rates in spikes per second for all downstream analyses. All three spike windows were analyzed per stimulus type to capture how subregion encoding differed across distinct phases of the neural response.

3.2 Encoding

Encoding analyses characterized the structure of neural population representations across subregions and stimulus types. For all encoding analyses, neuron-by-trial matrices of firing rates from all sessions and mice were concatenated along the neuron axis to form a single **population matrix** per subregion, stimulus type, and spike window.

3.2.1 Principal Component Analysis

Principal Component Analysis (PCA) (Pedregosa et al., 2018) was applied separately for each combination of subregion, stimulus type, and spike window. PCA provides an interpretable linear decomposition of population variance without requiring labeled data, making it well suited for exploratory characterization of representational manifolds across subregions. Input matrices were mean-centered per neuron before PCA. PCA figures show all trials projected onto the first two principal components, with points colored by stimulus identity.

3.2.2 Participation Ratio

Participation ratio (PR) quantifies effective dimensionality, calculated from the explained-variance spectrum where λ are the eigenvalues from PCA (Recanatesi et al., 2022):

$$\text{Participation Ratio (PR)} = \frac{(\sum \lambda)^2}{\sum \lambda^2}$$

The PR summarizes how many principal components carried meaningful variance, ranging from 1 (all variance concentrated in a single component) to n (variance equally distributed across all n components), with higher values indicating more distributed representations. It is sensitive to the full shape of the variance spectrum, making it a comparable metric across subregions and stimulus types. Since PR scales with neuron count, population matrices were subsampled to a fixed neuron count before analysis (278 for non-speech stimuli, 99 for speech) and formal statistical comparisons of PR values across subregions were therefore not possible. Scree plots showing the full variance spectrum and PR for each subregion, stimulus type, and spike window are provided in Supplemental Figures 2–5.

3.3 Decoding

3.3.1 Ridge Regression

Ridge regression (Hoerl and Kennard, 1970) decoding was applied to PT and AM stimuli only. Natural sounds and speech syllables have no inherent acoustic ordering, so assigning a numeric regression target would impose an arbitrary ranking that does not reflect any continuous acoustic dimension. Speech syllables are also defined along two independent and orthogonal acoustic dimensions (VOT and FT) and ridge regression would collapse this 2D space onto an arbitrary ordering that does not preserve the structure of the stimuli. For both natural sounds and speech, discriminability analysis (Section 3.3.2) was better suited.

Ridge regression was applied at the session level, independently for each subregion, stimulus type, and spike window. L2 regularization shrinks coefficients toward zero without zeroing any out, preserving contributions from the full population and reducing overfitting in high-dimensional correlated data. Session matrices for each subregion and stimulus type were **z-scored** independently and subsampled 100 times to 30 neurons per session to allow fair cross-

region comparisons. The regression target was the log-transformed stimulus parameter value, consistent with the logarithmic spacing of stimuli. Each subsample was evaluated with shuffled **5-fold cross-validation**, with alpha selected on training folds using RidgeCV over a grid of 200 logarithmically spaced values from 10^{-10} to 10^5 (see Fig. S6, S7) (Pedregosa et al., 2018). Session accuracy was evaluated using mean R^2 value computed on held-out testing data and averaged across subsamples (see Fig. S8).

3.3.2 Discriminability

Discriminability analyses quantify how distinguishable population responses to stimulus pairs are. Using **linear Support Vector Machine (SVM) classifiers** (Pedregosa et al., 2018), pairwise discriminability was computed on population matrices formed by concatenating neurons across all sessions and animals within each subregion and stimulus type. Predictor matrices were z-scored and subsampled to equal-neuron counts (278 per subregion for non-speech; 99 for speech) (Pedregosa et al., 2018). The regularization hyperparameter C was tuned over a grid of 20 logarithmically spaced values from 10^{-5} to 10^4 for each combination of sound type, subregion, and spike window (see Fig. S9–S13). Pairwise classification accuracy was estimated using shuffled **leave-one-out cross-validation** (Weese et al., 2025).

3.3.3 Statistical Analysis

Statistical comparisons were applied to summary distribution figures, focusing on the subregion comparisons central to the research questions. Neurons were randomly subsampled to a fixed count before computing any summary statistic, ensuring that differences in distribution reflected differences in neural representation rather than population size (see Fig. S1). All pairwise subregion comparisons within each sound type and spike window were evaluated using unpaired **Mann-Whitney U tests** (Mann and Whitney, 1947), corrected for multiple

comparisons using **Bonferroni correction** (Dunn, 1961). Significance was annotated on figures using bracket-and-star notation (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

3.4 Code and Reproducibility

All analyses and code for reproducing figures were implemented in Python and are publicly available at <https://github.com/itszoetom/AudCompAnalysis> . The repository is organized into analysis-specific folders with shared utilities for parameters, cross-validation helpers, and statistical annotations respectively. Random subsampling and cross-validation procedures used fixed or deterministic seeds where applicable to support reproducibility across runs.

4. Results

4.1 Encoding

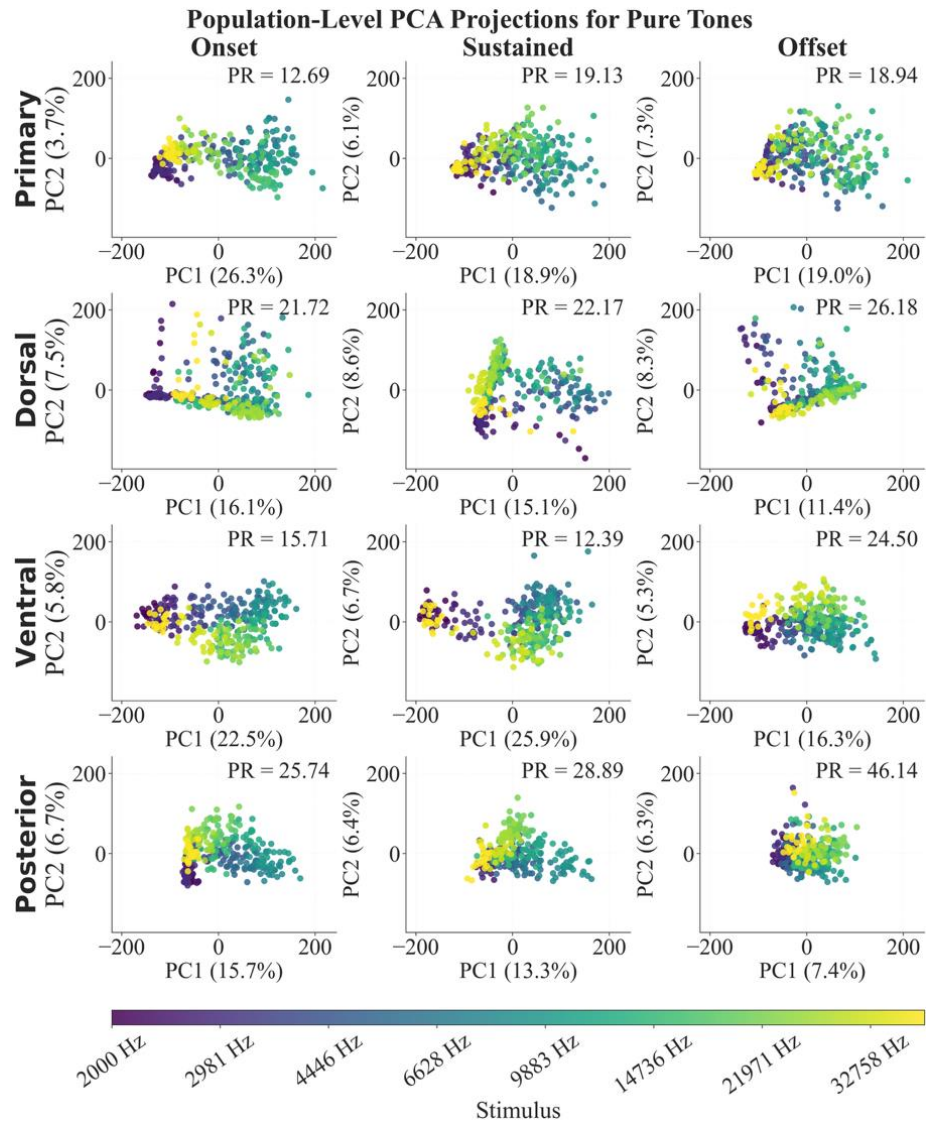


Figure 3: Pure Tones Neural Manifolds Across Subregions

Two-dimensional PCA projections of population firing rates for pure tones, shown for all four subregions (AudP, AudD, AudV, AudPo) and all three analysis windows (onset, sustained, offset). Each point represents one trial, colored by stimulus frequency. Participation ratio (PR) is annotated in each panel; higher PR indicates more distributed variance. Neuronal populations were subsampled to 278 per subregion, pooled across sessions, so statistical comparisons of PR values across subregions were not possible.

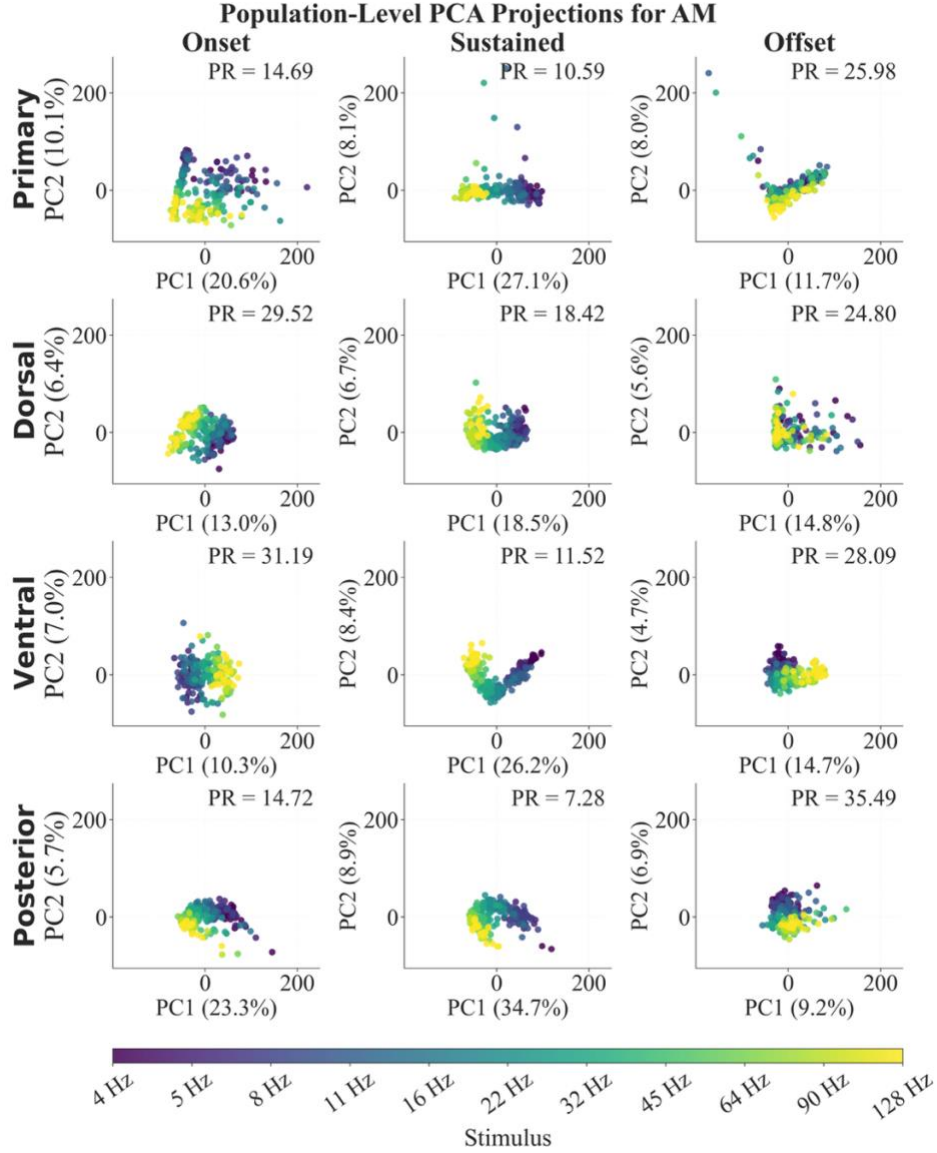


Figure 4: AM White Noise Neural Manifolds Across Subregions

Two-dimensional PCA projections of population firing rates for AM white noise, shown for all four subregions (AudP, AudD, AudV, AudPo) and all three analysis windows (onset, sustained, offset). Each point represents one trial, colored by stimulus modulation rate. Participation ratio ($PR = (\sum \lambda)^2 / \sum \lambda^2$) is annotated in each panel; higher PR indicates more distributed variance.

Neuronal populations were subsampled to 278 per subregion, pooled across sessions, so statistical comparisons of PR values across subregions were not possible.

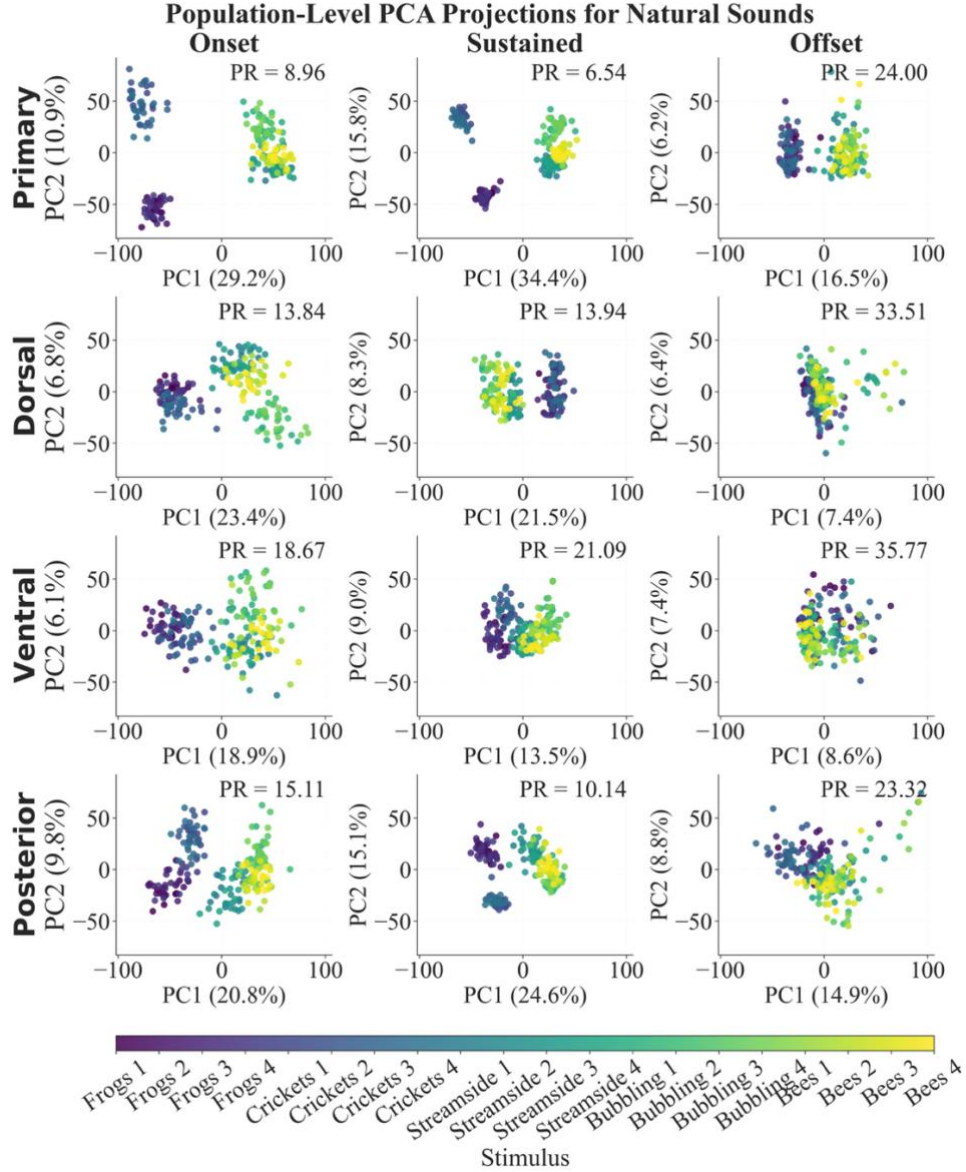


Figure 5: Natural Sounds Neural Manifolds Across Subregions

Two-dimensional PCA projections of population firing rates for natural sounds, shown for all four subregions (AudP, AudD, AudV, AudPo) and all three analysis windows (onset, sustained, offset). Each point represents one trial, colored by stimulus sound category and exemplar. Participation ratio ($PR = (\sum \lambda)^2 / \sum \lambda^2$) is annotated in each panel; higher PR indicates more distributed variance. Neuronal populations were subsampled to 278 per subregion, pooled across sessions, so statistical comparisons of PR values across subregions were not possible.

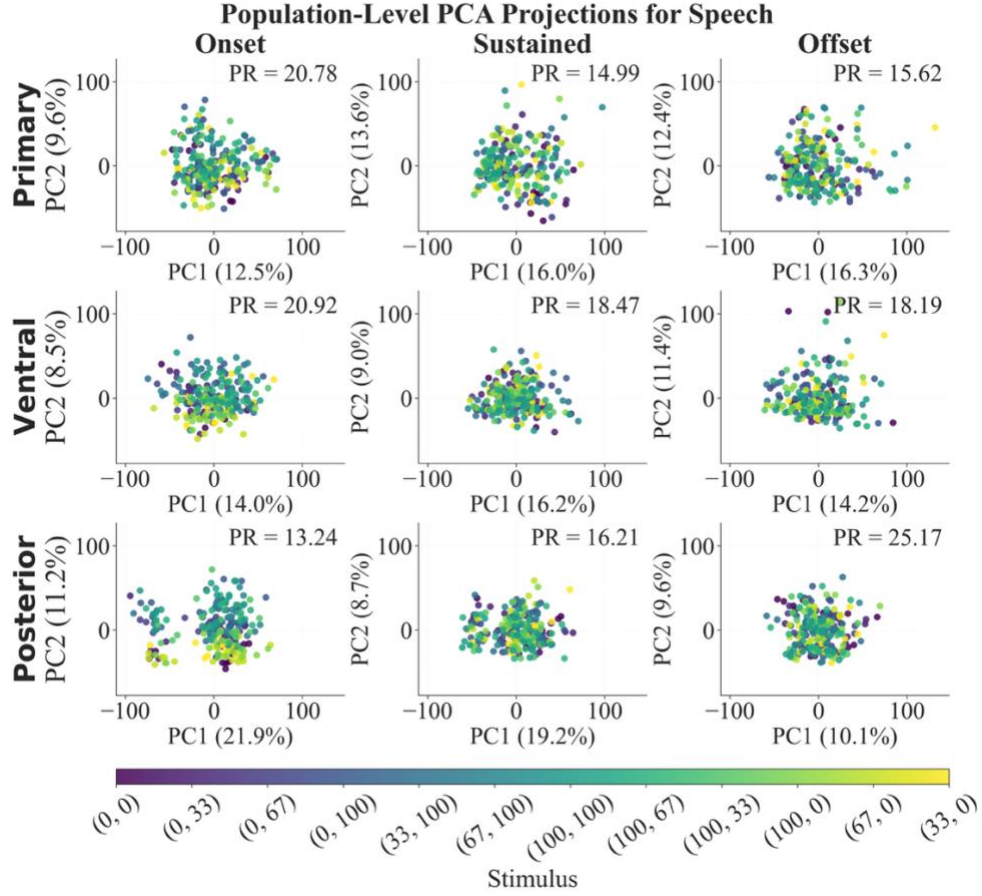


Figure 6: Speech Syllables Neural Manifolds Across Subregions

Two-dimensional PCA projections of population firing rates for speech syllables, shown for three subregions (AudP, AudV, AudPo; AudD excluded due to insufficient neuron count) and all three analysis windows (onset, sustained, offset). Each point represents one trial, colored by stimulus identity (VOT and FT coordinates). Participation ratio ($PR = (\sum \lambda)^2 / \sum \lambda^2$) is annotated in each panel; higher PR indicates more distributed variance. Neuronal populations were subsampled to 99 per subregion, pooled across sessions, so statistical comparisons of PR values across subregions were not possible.

To understand how sound information is organized across subregions, PCA was applied to population firing rate matrices, projecting neural responses onto axes of greatest variance to visualize the manifolds stimulus representations. PCA projections revealed that all four AC subregions formed structured **neural manifolds** for PT, AM, and natural sounds (Figs. 3–6).

For PT, trial projections along PC1 followed an ordered continuum, with points arranged from low to high frequency (Fig. 3). This tonotopic gradient was visible in all subregions during both onset and sustained windows. AudPo showed the highest PR values in the sustained and offset windows, with a notably high offset PR of 46.14 compared to a range of 18.94 to 26.18 across the other subregions, indicating more distributed PT representations in later response phases.

For AM white noise, manifolds showed curved arc structure during onset and sustained windows across all subregions, reflecting an ordered representation of modulation rate (Fig. 4). This geometry largely collapsed in the offset window, suggesting that later-phase AM representations are lower-dimensional. AudP showed structured arc geometry across onset and sustained windows. AudV showed a notably lower sustained PR (11.52) compared to AudD (18.42) and AudPo (14.72), with arc structure that was less defined in the sustained window, indicating more compressed AM representations. AudD showed the weakest arc structure overall despite its higher sustained PR, consistent with that variance being unstructured rather than reflecting ordered modulation rate representations.

For natural sounds, all subregions showed categorical clustering in the PCA projections, with exemplars from the same environmental category grouping together in the projection space (Fig. 5). AudP and AudPo showed compact, lower-dimensional representations with comparatively low sustained PR values, and AudPo showed cluster separation comparable to or stronger than AudP across windows. AudD showed sustained categorical clustering with a PR of 13.94, more distributed than AudP or AudPo but with visible category structure in the projections. AudV showed the weakest categorical clustering and the highest sustained PR of

21.09, indicating that natural sound representations were more spread across dimensions rather than organized by category.

For speech syllables, PCA projections showed overlapping, low-structure distributions across all three included subregions (Fig. 6). No subregion produced clearly separated syllable representations in the low-dimensional projection. Participation ratios were broadly similar across subregions in the sustained window, with AudPo showing a slightly higher onset and more visible trial clustering at onset compared to AudP and AudV. Given that neurons were pooled across sessions, formal statistical comparisons across subregions on PR values were not possible. Taken together, PCA results show that subregion neural manifolds vary with stimulus type, with AudV showing the weakest categorical structure for natural sounds and AudPo showing comparatively stronger structure for complex sounds at onset.

4.2 Decoding

4.2.1 Ridge Regression

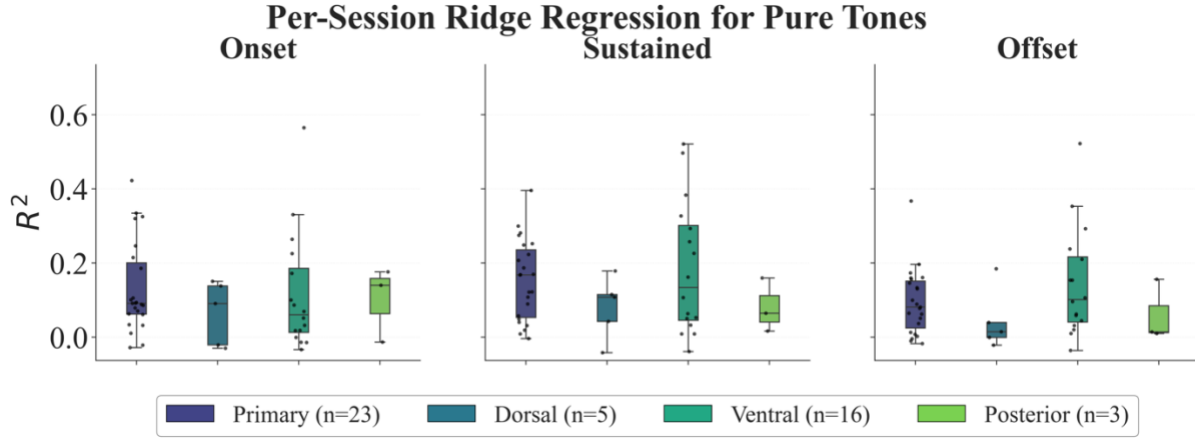


Figure 7: Pure Tones Per-Session Ridge Regression R^2 Across Subregions

Per-session ridge regression R^2 scores for pure tones across AC subregions and spike windows. Each boxplot summarizes R^2 across sessions and 100 random subsamples of 30 neurons per session. The regression target was log-transformed frequency. No pairwise Mann-Whitney U comparisons reached significance after Bonferroni correction.

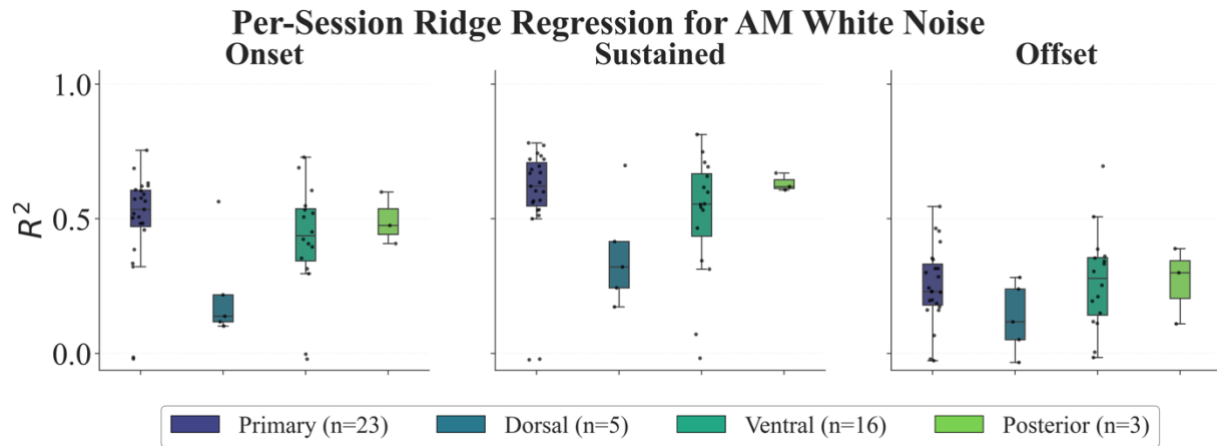


Figure 8: AM Per-Session Ridge Regression R^2 Across Subregions

Per-session ridge regression R^2 scores for AM white noise across AC subregions and spike windows. Each boxplot summarizes R^2 across sessions and 100 random subsamples of 30 neurons per session. The regression target was log-transformed modulation rate. No pairwise Mann-Whitney U comparisons reached significance after Bonferroni correction.

To quantify how accurately acoustic stimulus parameters could be predicted from population responses, ridge regression was applied at the session level, predicting log-transformed stimulus values from z-scored neural responses using 5-fold cross-validation (Figs. 7–8). For PT, AudP and AudV showed the strongest median per-session R^2 in the sustained and offset windows. AudD and AudPo showed lower median R^2 values across all windows, with the gap smallest at onset and growing toward offset. Pairwise Mann-Whitney U tests with Bonferroni correction revealed no significant differences, though statistical difference between AudV and AudD approached significance prior to correction ($p < 0.1$).

For AM white noise, subregion differences were more pronounced. AudP, AudV, and AudPo all showed higher median R^2 than AudD across the onset and sustained windows, with AudD again showing the weakest performance across all windows. Pairwise Mann-Whitney U tests with Bonferroni correction revealed no significant differences, though the comparison between AudP and AudD approached significance prior to correction ($p < 0.1$). Together, these results show that AudP and AudV most consistently predicted simple spectral and temporal stimulus parameters from population responses, while AudD showed the weakest prediction performance across both stimulus types and windows.

4.2.2 Discriminability

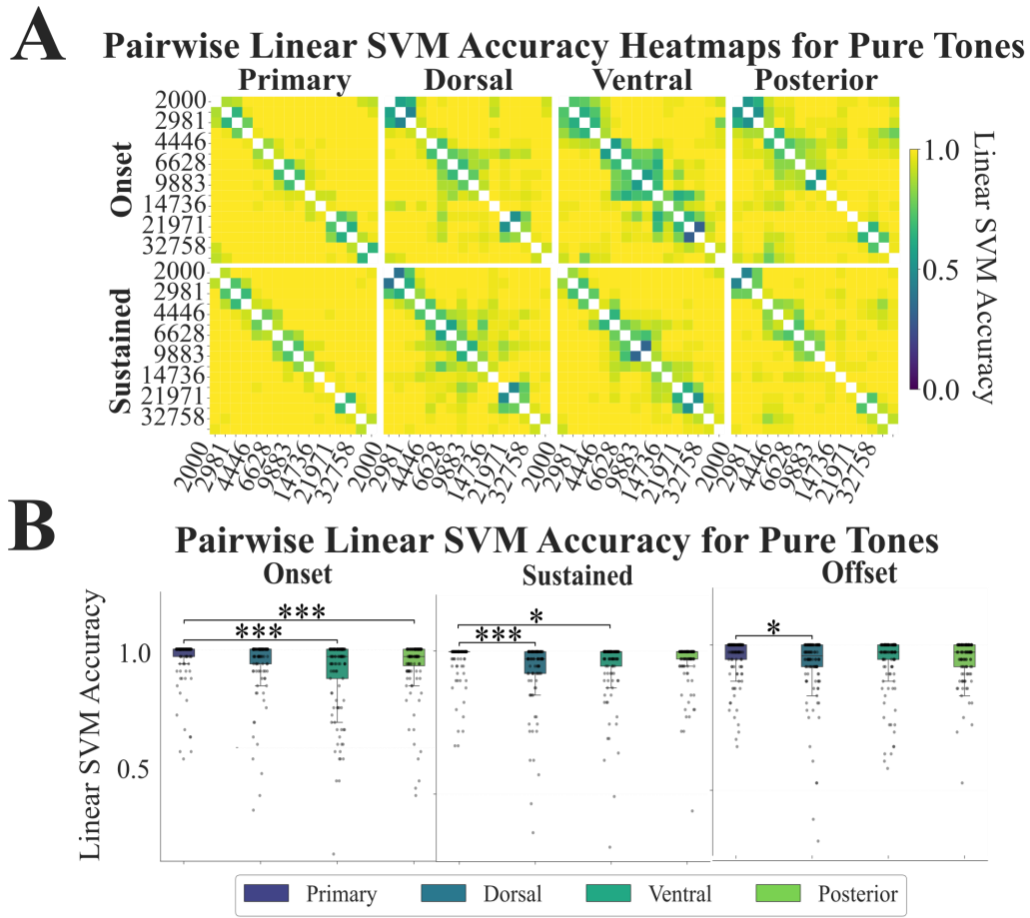


Figure 9: Pure Tones Pairwise Linear SVM Discriminability Across Subregions

Pairwise linear SVM classification accuracy for PT stimuli. (A) PT pairwise accuracy heatmaps by subregion and spike window; rows and columns index individual frequencies and color indicates leave-one-out classification accuracy. (B) PT accuracy distributions by subregion and spike window. Significance brackets indicate pairwise Mann-Whitney U tests with Bonferroni correction (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Neuron counts were subsampled to 278 per subregion before analysis.

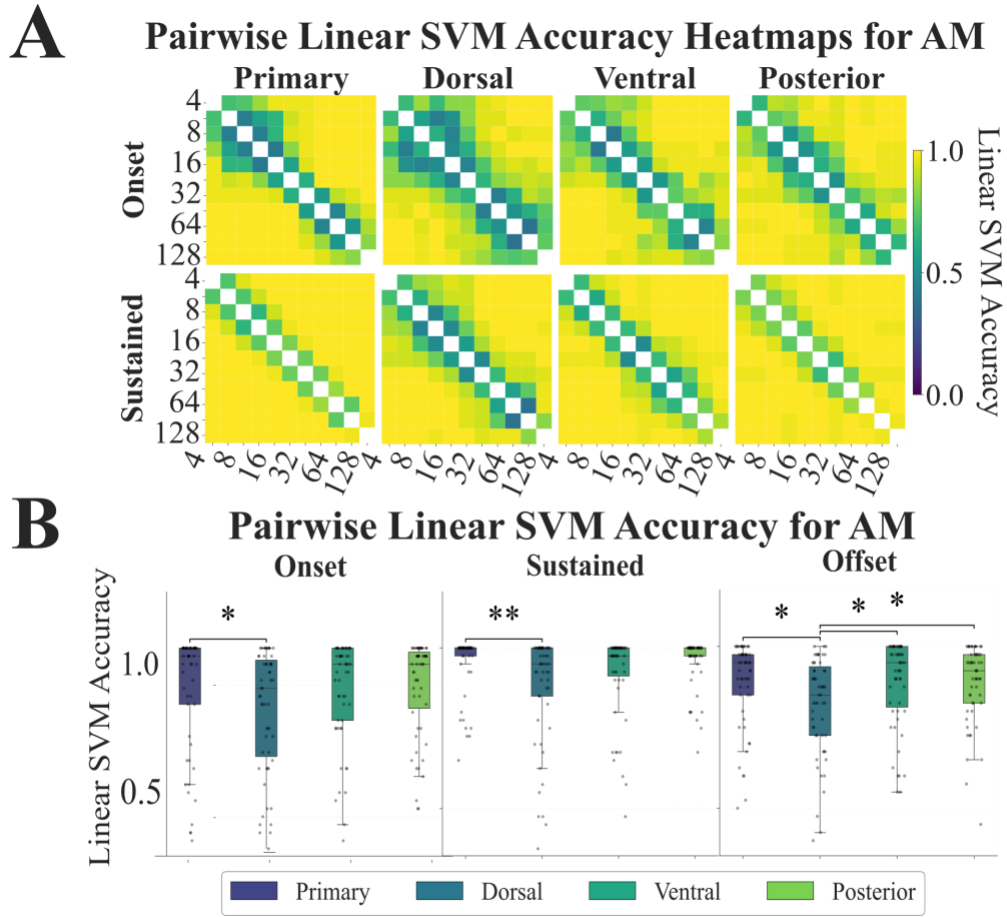
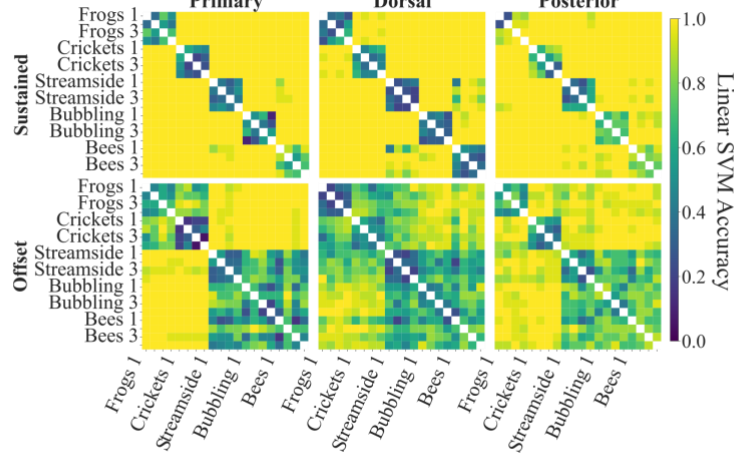


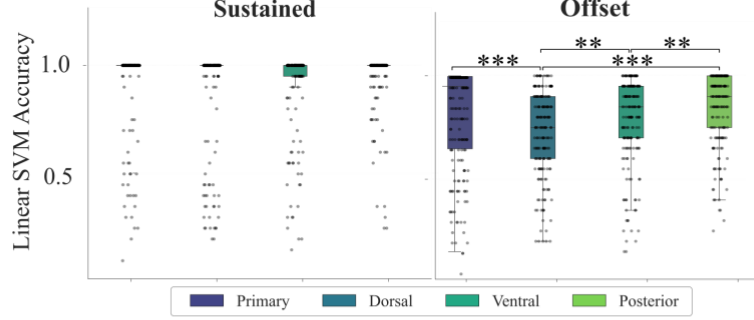
Figure 10: AM Pairwise Linear SVM Discriminability Across Subregions

Pairwise linear SVM classification accuracy for AM stimuli. (A) AM pairwise accuracy heatmaps by subregion and spike window; rows and columns index individual modulation rates and color indicates leave-one-out classification accuracy. (B) AM accuracy distributions by subregion and spike window. Significance brackets indicate pairwise Mann-Whitney U tests with Bonferroni correction (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Neuron counts were subsampled to 278 per subregion before analysis.

A Pairwise Linear SVM Accuracy Heatmaps for Natural Sounds



B Pairwise Linear SVM Accuracy for Natural Sounds



C Within vs. Between Category

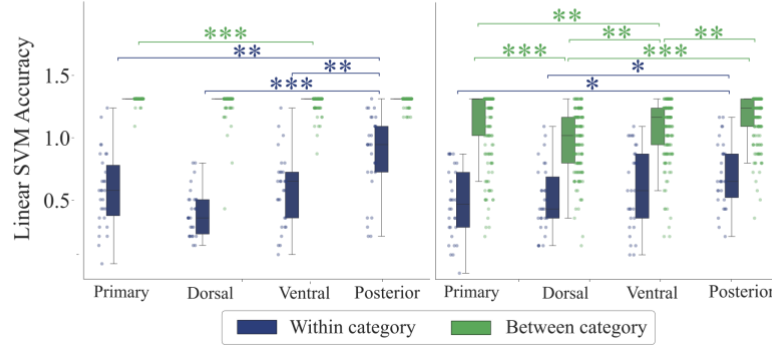


Figure 11: Natural Sounds Pairwise Linear SVM Discriminability Across Subregions

Pairwise linear SVM classification accuracy for natural sounds. (A) Natural sounds pairwise accuracy heatmaps by subregion and spike window; rows and columns index individual stimuli and color indicates leave-one-out classification accuracy. (B) Natural sounds accuracy distributions across all unique stimulus pairs. (C) Natural sounds within-category versus between-category accuracy distributions. Significance brackets indicate pairwise Mann-Whitney U tests with Bonferroni correction (* p < 0.05, ** p < 0.01, *** p < 0.001). Neuron counts were subsampled to 278 per subregion before analysis.

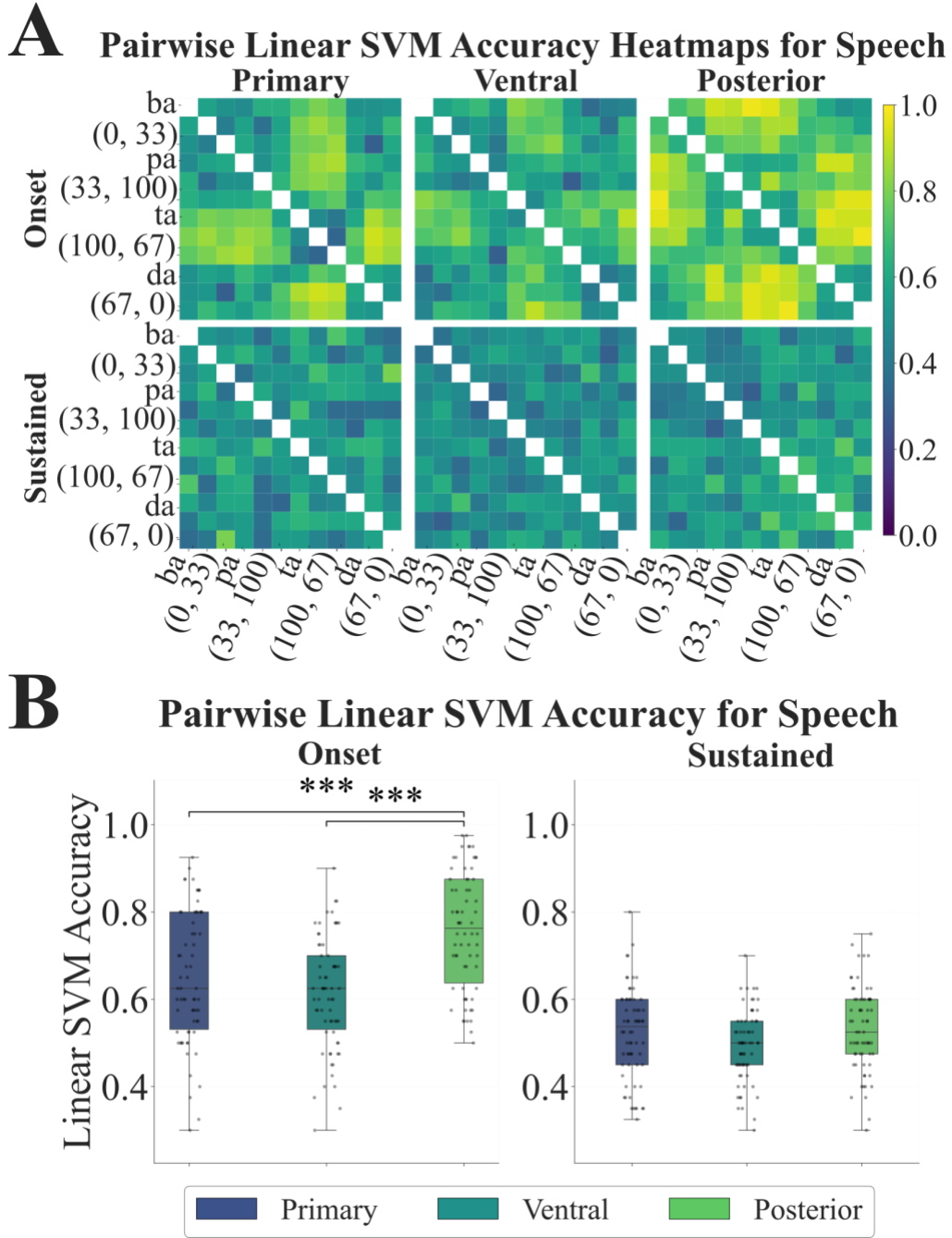


Figure 12: Speech Syllable Pairwise Linear SVM Discriminability Across Subregions

Pairwise linear SVM classification accuracy for speech syllable stimuli. (A) Speech syllable pairwise accuracy heatmaps by subregion and spike window. (B) Speech syllable accuracy distributions by subregion and spike window. Significance brackets show pairwise Mann-Whitney U tests with Bonferroni correction (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Neuron counts were subsampled to 99 per subregion before analysis.

To assess how well population responses could distinguish between individual stimuli, pairwise linear SVM classifiers were fit to population matrices pooled across sessions within each subregion, estimating discriminability as leave-one-out classification accuracy for each stimulus pair (Fig. 9). Pairwise discriminability was strong for both PT and AM across most subregions, with most stimulus pairs classified well above chance. For PT, all four subregions showed strong off-diagonal structure in the heatmaps and median pairwise accuracy above chance across windows. AudD and AudV showed the weakest PT discriminability overall, with significantly lower pairwise accuracy than AudP at onset and sustained ($p < 0.001$). AudPo showed discriminability comparable to AudP across windows.

For AM, discriminability was strong across AudP, AudV, and AudPo, with all three performing comparably. AudD was significantly weaker than the other three subregions at all three spike windows ($p < 0.05$), with reduced accuracy most pronounced for pairs of similar modulation rates, visible as reduced accuracy near the diagonal of the AudD heatmap. This pattern was consistent with the absence of structured arc geometry in AudD PCA projections and with its lower AM ridge regression R^2 values across all windows.

For natural sounds, all subregions showed clear block-diagonal structure in the pairwise heatmaps, indicating that between-category pairs were easier to discriminate than within-category pairs across all subregions and windows (Fig. 10). AudPo showed the strongest within-category discriminability, with significantly higher within-category accuracy than all other subregions in the sustained window ($p < 0.05$) and the highest values at onset and offset. AudV showed the weakest discriminability overall, with lower accuracy than AudP and AudPo across windows. AudD showed moderate discriminability, performing above AudV in some windows but below AudP and AudPo overall.

For speech syllables, discriminability was strongest in the onset window across all three subregions (Fig. 11). AudPo showed the highest onset accuracy, significantly above both AudP and AudV ($p < 0.001$), with heatmaps showing structured separability particularly for syllable pairs differing in both VOT and FT. AudP and AudV performed comparably at onset and sustained windows, with no significant difference between them. Discriminability largely collapsed in the sustained and offset windows across all subregions. Taken together, these discriminability results show that AudPo exhibits the strongest pairwise discrimination for complex sounds, AudV performs comparably to AudP for simple stimuli but is the weakest subregion for natural sounds and speech, and AudD shows consistently reduced discriminability across all stimulus types.

5. Discussion

Encoding analyses of population neural manifolds and dimensionality, and decoding analyses of stimulus prediction and discriminability, together reveal functional specialization in AC subregions. Each subregion showed a distinct representational profile, and the relative contribution of each area shifted depending on the acoustic content of the stimulus. These patterns are most consistent with a distributed processing framework rather than a fixed hierarchy or parallel independent channels, with implications for understanding the functional organization of secondary AC in the mouse.

The dataset size, in both sessions per subregion and neurons per session, limited the results of both analyses. With as low as three sessions per subregion, no ridge regression comparisons reached significance, though several approached the corrected threshold, suggesting subregion differences that larger session counts may resolve. Low neuron counts per session required pooling across sessions for PCA, preventing formal statistical comparisons of participation ratio values across subregions. Larger datasets with more sessions and neurons per subregion would directly address both limitations. Further studies could analyze cross-region connectivity and whether the distributed processing pattern reflects coordination across subregions or independent contributions from each. Expanding the stimulus set to include spatially varied stimuli would test AudD's proposed role in sound localization directly, which the present stimulus set could not address.

AudP is the most well-characterized AC subregion and serves as the functional baseline against which the secondary subregions are compared here. Across all stimulus types and analyses, AudP showed strong and consistent performance, including well-formed manifolds for PT, AM and natural sounds in PCA, strong ridge regression prediction, and high pairwise

discriminability across all sound types. This breadth of performance is consistent with the established role of primary AC in spectrotemporal feature processing (Hackett, 2011), and provides a reliable reference point for interpreting differences in the secondary subregions.

AudV differed from AudP specifically as stimulus complexity increased. For simple sounds, AudV performed comparably to AudP across PCA arc geometry, ridge regression performance, and AM discriminability, indicating robust encoding of spectral and temporal features. PT discriminability was weaker than AudP, suggesting AudV's population carries less fine-grained frequency information despite similar overall prediction accuracy, but it remained stronger than AudD and AudPo overall. For natural sounds and speech, AudV was the weakest subregion across both PCA and linear SVM discriminability, showing the most distributed representations and the lowest pairwise accuracy. These results suggest AudV preference for lower-complexity acoustic features.

AudPo showed a distinct specialization for complex sound processing. For PT, AudPo showed weaker PCA manifold structure and ridge regression performance compared to AudP, indicating limited engagement with simple spectral features. For AM and natural sounds, AudPo performed more strongly. AM manifolds showed clear arc structure, and decoding results were comparable to AudP and AudV. For natural sounds, AudPo showed compact categorical clustering in PCA and the strongest within-category discriminability of all subregions in the sustained window, significantly above all others. AudPo also showed the highest speech onset discriminability, significantly above AudP and AudV, consistent with previous reports of selectivity to VOT and FT in the posterior region of mice (Mohn et al., 2024). Taken together, AudPo's weak simple sound performance and specialization for more complex stimuli is consistent with findings that non-primary auditory areas support stronger discrimination of

complex sounds (Feigin et al., 2021), and suggests a preferential role for higher-order acoustic processing.

AudD showed consistently weak performance across all stimulus types and analyses. For PT, ridge regression R^2 values were visibly lower than AudP and AudV across all windows, and pairwise discriminability was significantly lower than AudP at onset. For AM, AudD lacked structured arc geometry in PCA and showed the lowest ridge regression R^2 values across all windows. For natural sounds, AudD showed moderate categorical clustering in PCA but weaker pairwise discriminability than AudP and AudPo. Rather than reflecting a preference for a particular acoustic feature type, AudD's profile suggests a comparatively limited role in representing the acoustic features tested here. One prominent hypothesis proposes that AudD is specialized for sound localization, forming part of an auditory "where" pathway analogous to the dorsal visual stream (Rauschecker and Tian, 2000). In cats, bilateral deactivation of the dorsal auditory zone impairs sound localization behavior (Stecker et al., 2005), and a double dissociation between spatial and pattern processing has been demonstrated in separate posterior and anterior auditory fields (Lomber and Malhotra, 2008). The weak performance observed across AudD may reflect a mismatch between these stimuli and AudD's functional role rather than a general processing deficit.

Taken together, these results most strongly support a distributed processing framework, in which each subregion shows a preference for different acoustic features rather than all areas responding equivalently to every sound. Each subregion showed a distinct specialization profile that emerged depending on the complexity and structure of the sound being heard. AudP showed broad and consistent engagement across all sound types, with AudV contributing most for simple spectral and temporal features and AudPo taking a stronger role for complex categorical sounds.

AudD contributed minimally across all conditions. A parallel processing model is inconsistent with these results because it predicts each secondary subregion should hold a fixed, exclusive specialization that does not change with sound type or complexity. The results instead showed flexible, stimulus-dependent specializations that shifted with acoustic content, and no subregion was entirely unresponsive to any stimulus type.

A hierarchical model predicts that representational complexity increases monotonically from primary to higher-order areas (Hackett, 2011; Joachimsthaler et al., 2014). The results offer meaningful partial support for this, as AudPo outperformed other subregions for natural sounds and speech, and the pattern of increasing complex sound performance from AudP to AudPo is consistent with a processing gradient from primary to higher-order areas. AudV's stronger performance for simple sounds relative to complex ones also fits a hierarchy in which secondary areas take on more specialized roles. However, AudD's weak performance across all stimulus types does not fit a strict hierarchical sequence, as a higher-order area would be expected to show at least selective engagement with complex stimuli. The distributed model accommodates all of these patterns, including the stimulus-dependent shifts in subregion contributions, the partial gradient toward AudPo for complex sounds, and AudD's limited role, without requiring fixed processing order or equal participation across subregions (Hackett, 2011).

6. Conclusion

This study analyzed how four regions of the AC in mice respond to sounds of increasing complexity, from simple tones to natural sounds and speech. The most studied region, AudP, responded reliably across all sound types and served as a stable reference for comparing the other regions. AudV performed comparably to AudP for simple sounds but was the weakest region for natural sounds and speech, suggesting it is better suited to processing basic acoustic features. AudPo showed the opposite pattern, contributing little for simple sounds but showing the strongest responses to complex sounds, including the best separation of natural sound categories and speech of all regions tested. AudD showed consistently limited responses across all sound types, suggesting its role may involve acoustic features not tested here. Taken together, these results are most consistent with a distributed processing model, in which different regions contribute selectively depending on the acoustic content of the sound being heard. The data offer partial support for a hierarchical gradient, as AudV performed more strongly for simple sounds and AudPo for complex ones, which is consistent with the idea that higher-order regions take on a greater role as acoustic demands increase. AudD's limited responses across all conditions do not fit this pattern and may reflect specialization for a feature outside the scope of the present stimulus set, such as sound localization.

These results have broader relevance to conditions in which the brain's ability to process sound breaks down. If regions like AudPo are specifically important for speech and complex sounds, damage or dysfunction concentrated in those areas could impair speech understanding while leaving simpler hearing intact. For people with cochlear implants, which restore hearing at the ear but rely on the brain to interpret the signal, knowing which regions are most critical for complex sound processing may help explain why outcomes differ so much across patients. These

findings are specific to mice and would need to be replicated in animals more comparable to humans before drawing direct clinical conclusions, but they add to growing evidence that the AC functions as a coordinated system of specialized regions rather than a single processing center, with implications for how hearing loss and AC dysfunction are understood and treated.

7. Glossary

Acoustic content: The spectral and temporal features that characterize a sound, including frequency composition, amplitude envelope, and modulation structure.

Bonferroni correction: A multiple-comparisons correction that divides the significance threshold by the number of tests performed, controlling the family-wise error rate.

Dimensionality: The number of independent dimensions needed to describe variation in a dataset. High-dimensional data, such as firing rates across hundreds of neurons, can often be approximated in fewer dimensions if responses are correlated.

Discriminability: The degree to which population responses to two stimuli can be distinguished from one another. Measured here as pairwise classification accuracy using a linear SVM classifier: higher accuracy indicates that the two population responses occupy more separable regions of the response space, and therefore that the subregion encodes those stimuli as more distinct.

Firing rate: The number of action potentials (spikes) a neuron produces per unit time, expressed in spikes per second (Hz), calculated by counting spikes within a defined window and dividing by window duration.

5-fold cross-validation: A cross-validation strategy that partitions data into five equal folds, training on four and testing on one, rotating until all folds have served as the test set.

Neural manifold: The geometric structure formed by population responses when projected into a low-dimensional space using dimensionality reduction. A well-organized manifold indicates that neurons collectively represent stimulus identity in a structured, interpretable pattern.

Neuropixels: A type of high-density silicon recording probe developed for neuroscience that allows simultaneous extracellular recording from hundreds of neurons across multiple brain

regions. Used here to record population responses across AC subregions in awake head-fixed mice.

Participation ratio (PR): A summary statistic computed from the PCA variance spectrum that quantifies how many dimensions carry meaningful variance in a population response. Higher values indicate more distributed representations; lower values indicate variance concentrated in fewer dimensions.

Population responses: The firing rates of all simultaneously recorded neurons in a subregion on a given trial. Analyzed here as vectors in a high-dimensional space, one dimension per neuron, where each point represents the state of the population on a single trial.

Population matrix: A two-dimensional array of neural firing rates with neurons as rows and trials as columns, used as input to all population-level analyses. Each entry gives the mean firing rate of one neuron on one trial within a defined spike window.

Leave-one-out cross-validation (LOOCV): A cross-validation strategy in which one trial is held out as the test set and all remaining trials are used to train the classifier, repeated for every trial. Used here for pairwise SVM discriminability.

Linear Support Vector Machine Classifier: A linear classification model that finds the hyperplane maximizing the margin between classes. Used here for pairwise stimulus discriminability.

Mann-Whitney U test: A nonparametric two-sample test that assesses whether one distribution tends to produce larger values than another, without assuming normality.

PCA (Principal Component Analysis): A linear dimensionality-reduction method that reprojects data onto orthogonal axes of greatest variance, enabling visualization and quantification of population structure.

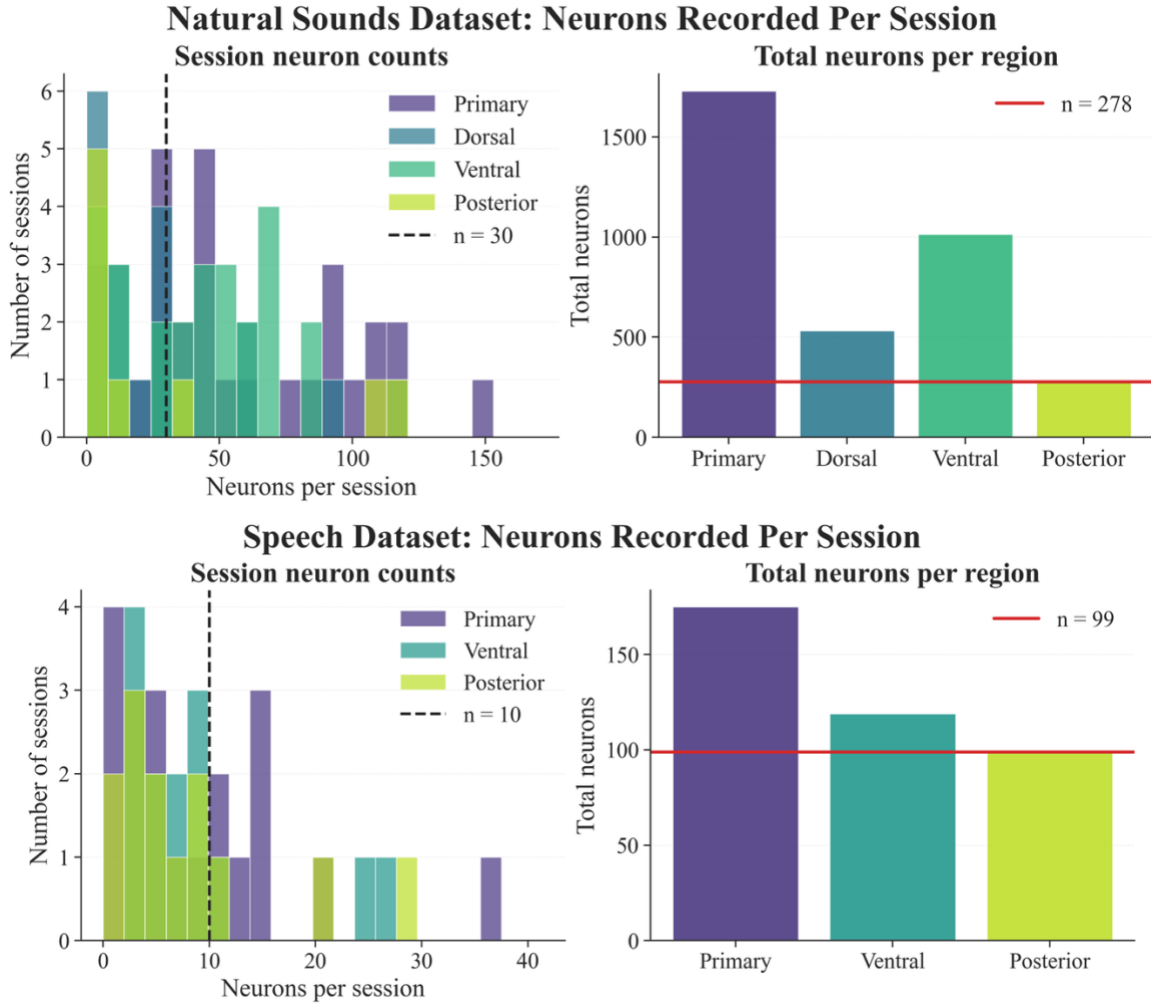
Ridge regression: A linear regression model with L2 regularization that penalizes large coefficients, reducing overfitting when predictors are high-dimensional or collinear.

Subsampling: The procedure of randomly selecting a fixed number of neurons from a larger recorded population before analysis, used here to ensure fair comparisons across subregions with different neuron counts.

Spike window: A defined time interval relative to stimulus onset within which spikes are counted to compute firing rate. Three windows are used per stimulus type: onset, sustained, and offset.

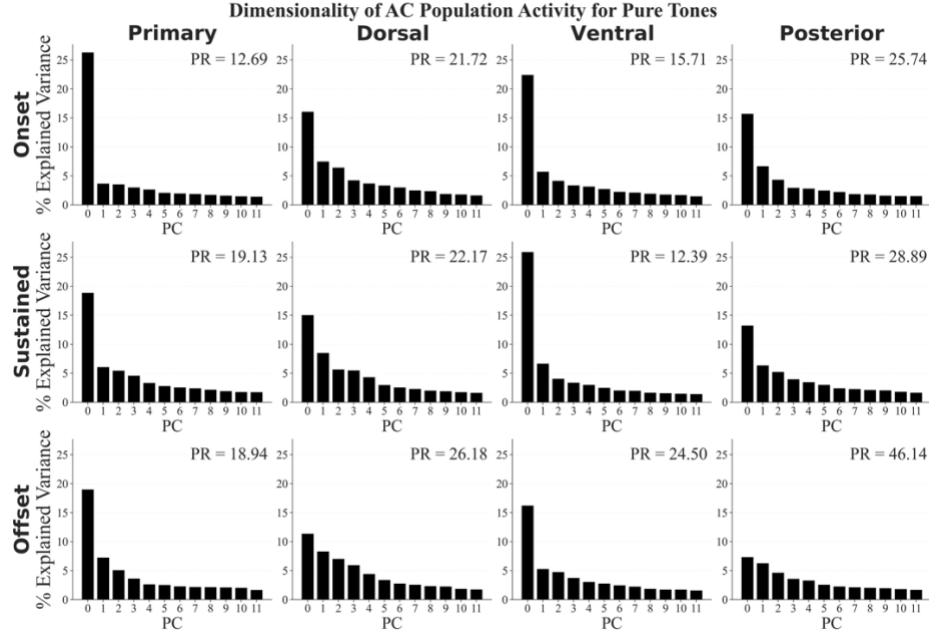
StandardScaler / Z-scoring: A preprocessing step that subtracts the mean and divides by the standard deviation of each feature, producing zero-mean unit-variance inputs for model fitting.

8. Appendix



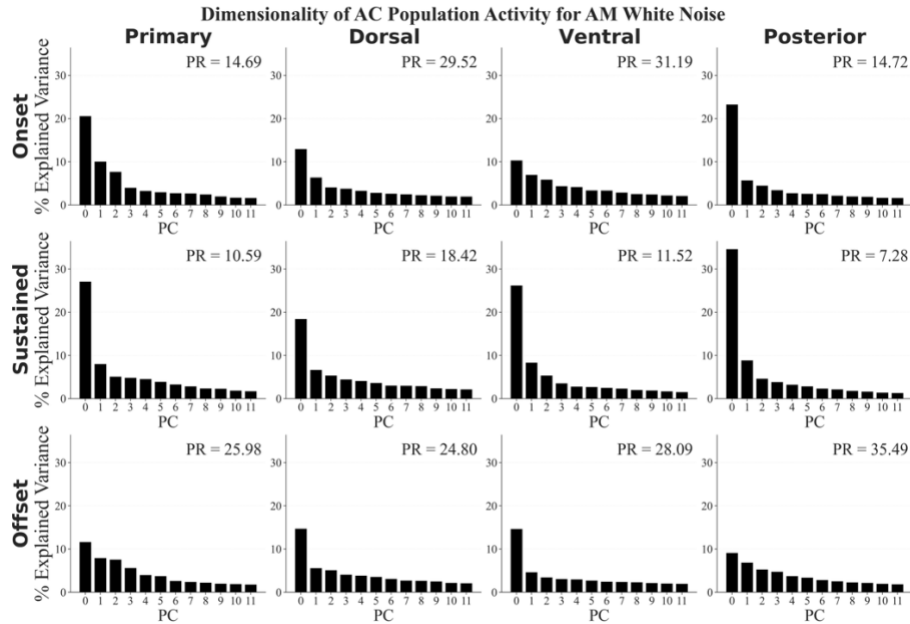
Supplemental Figure S1: Distribution of Neurons Per Session and Subregion

Histograms of neurons recorded per session (left) and total neurons per subregion (right) for the natural sounds dataset (AudP, AudD, AudV, AudPo) and speech dataset (AudP, AudV, AudPo). Each bar reflects the number of sessions with a given neuron count within that subregion. The thin dashed vertical line marks the per-session minimum for ridge regression analyses (30 neurons, natural sounds; 10 neurons, speech). The thick solid vertical line marks the population subsampling target for encoding and discriminability analyses (278 neurons, natural sounds; 99 neurons, speech); sessions below this threshold were excluded from population-level analyses. The unequal session counts per subregion motivate the fixed-count subsampling procedure described in Sections 3.2.1, 3.2.2, and 3.3.2. See Supplemental Figures 2–5 for the full variance spectra used in encoding analyses.



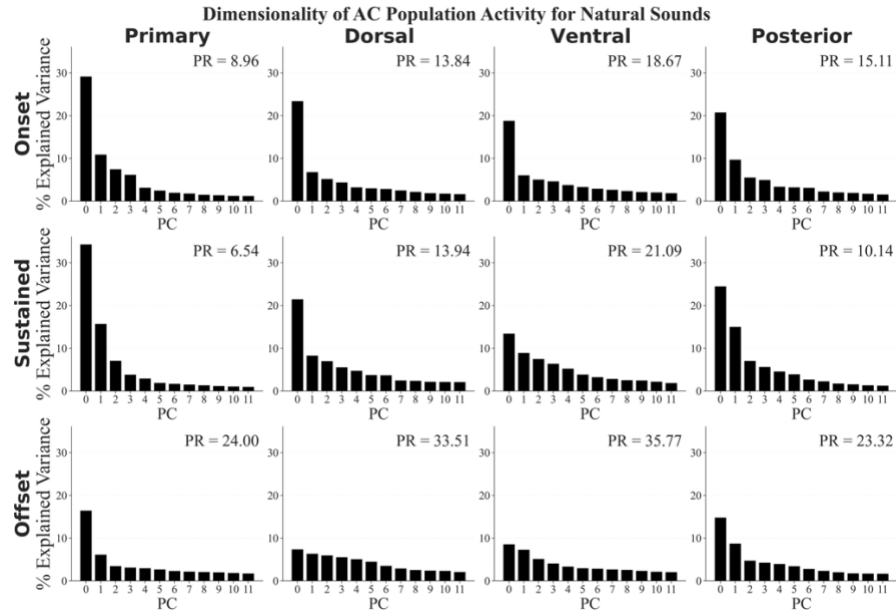
Supplemental Figure S2: Pure Tones PCA Scree Plots Per Subregion and Spike Window

Percent explained variance captured by each PC (1–12) for each combination of AC subregion and spike window. The participation ratio ($PR = (\sum \lambda)^2 / \sum \lambda^2$) is annotated in each panel. Neuron counts equalized to 278 per subregion.



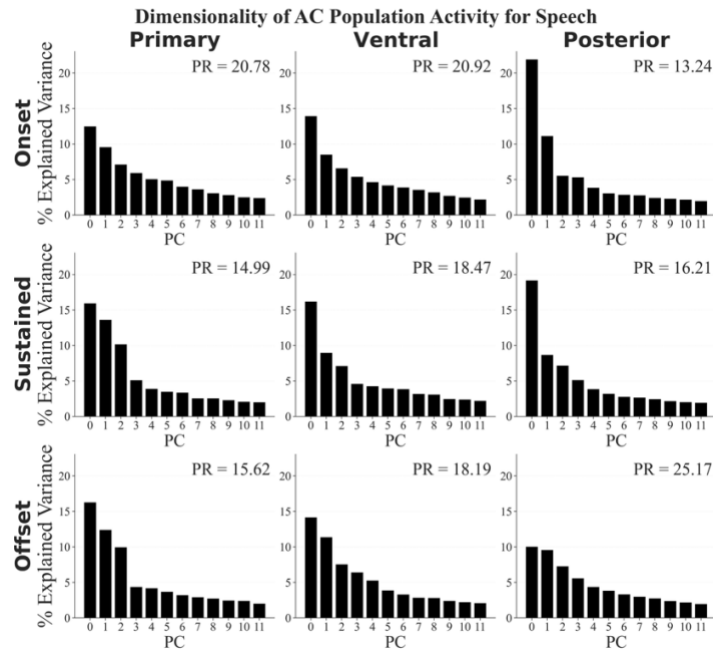
Supplemental Figure S3: AM PCA Scree Plots Per Subregion and Spike Window

Percent explained variance captured by each PC (1–12) for each combination of AC subregion and spike window. The participation ratio ($PR = (\sum \lambda)^2 / \sum \lambda^2$) is annotated in each panel. Neuron counts equalized to 278 per subregion.



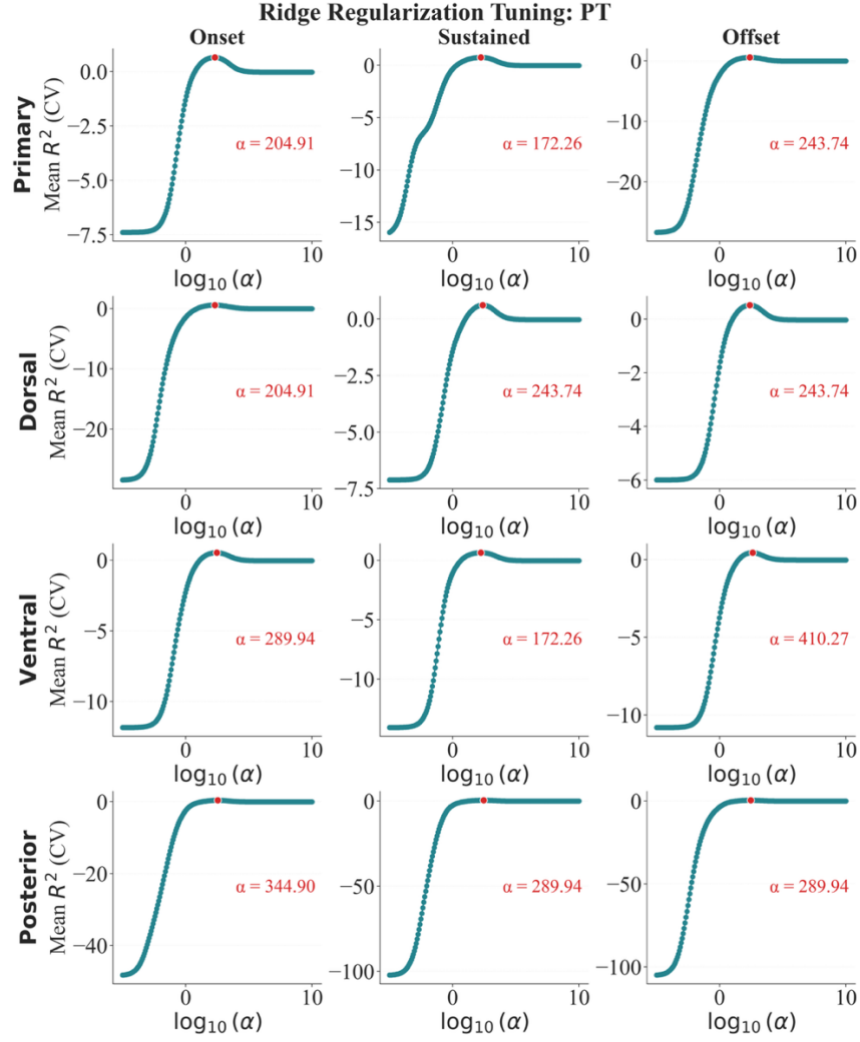
Supplemental Figure S4: Natural Sounds PCA Scree Plots Per Subregion and Spike Window

Percent explained variance captured by each PC (1–12) for each combination of AC subregion and spike window. The participation ratio ($PR = (\sum \lambda)^2 / \sum \lambda^2$) is annotated in each panel. Neuron counts equalized to 278 per subregion.



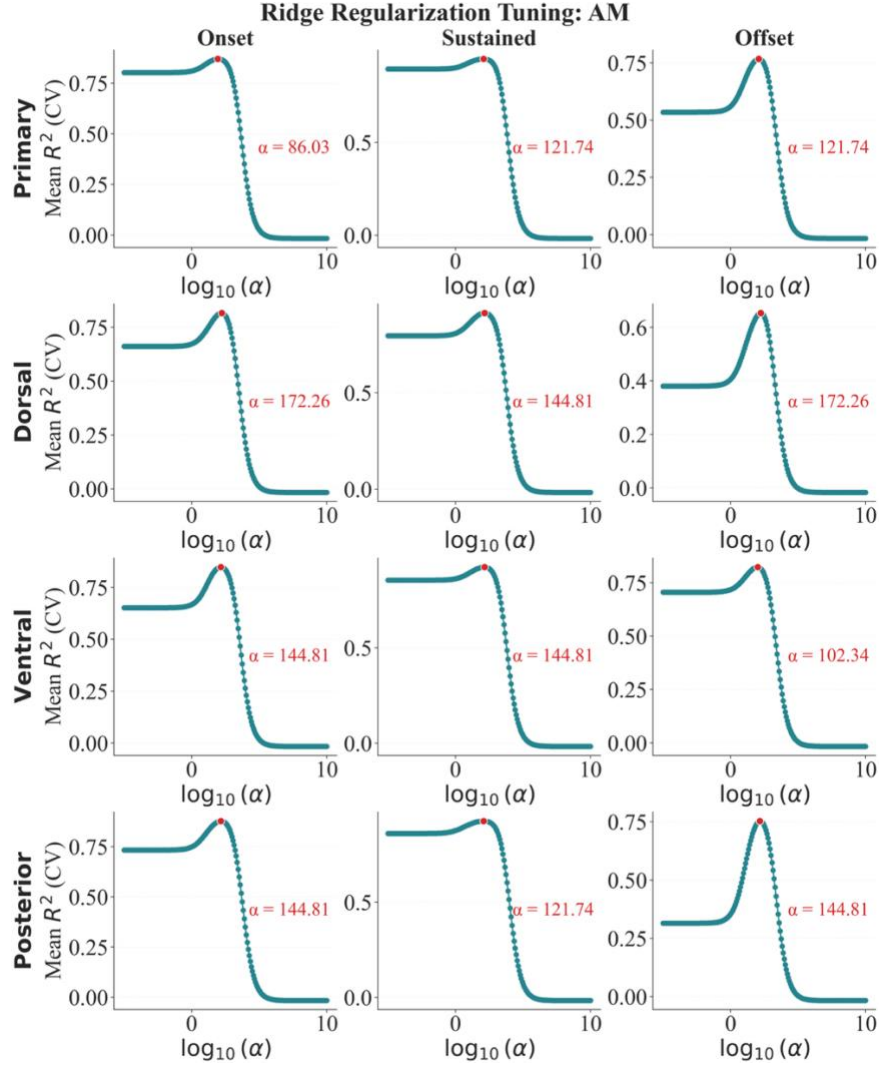
Supplemental Figure S5: Speech PCA Scree Plots Per Subregion and Spike Window

Percent explained variance captured by each PC (1–12) for each combination of AC subregion and spike window. The participation ratio ($PR = (\sum \lambda)^2 / \sum \lambda^2$) is annotated in each panel. Neuron counts equalized to 278 per subregion.



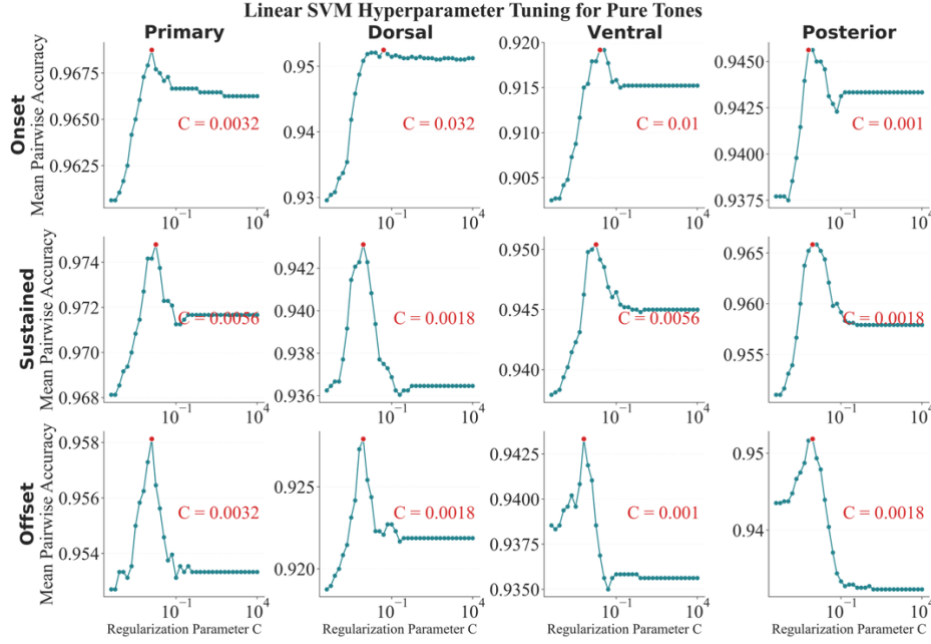
Supplemental Figure S6: Pure Tones Ridge Regression Alpha Tuning Curve

Mean ridge regression R^2 (5-fold CV) as a function of regularization parameter α ($\log_{10}$ scale) for each combination of auditory cortical subregion and spike window, for pure tone stimuli. The grid consisted of 200 logarithmically spaced values from 10^{-10} to 10^5 . The red vertical marker indicates the selected α value achieving the highest mean cross-validated R^2 on training folds. Alpha was selected independently for each session subsample.



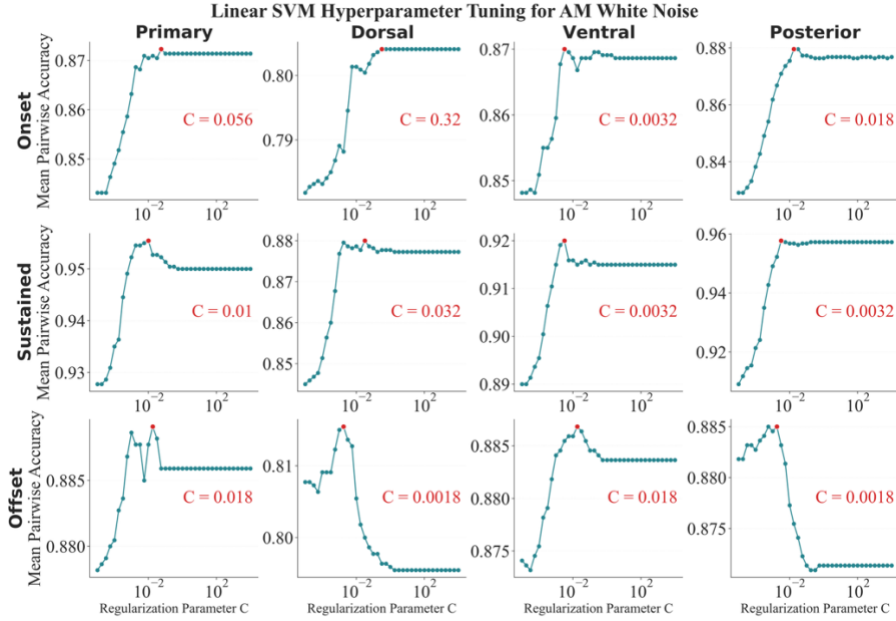
Supplemental Figure S7: AM Ridge Regression Alpha Tuning Curve

Mean ridge regression R^2 (5-fold CV) as a function of regularization parameter α ($\log_{10}$ scale) for each combination of auditory cortical subregion and spike window, for AM white noise stimuli. The grid consisted of 200 logarithmically spaced values from 10^{-10} to 10^5 . The red vertical marker indicates the selected α value achieving the highest mean cross-validated R^2 on training folds. Alpha was selected independently for each session subsample.



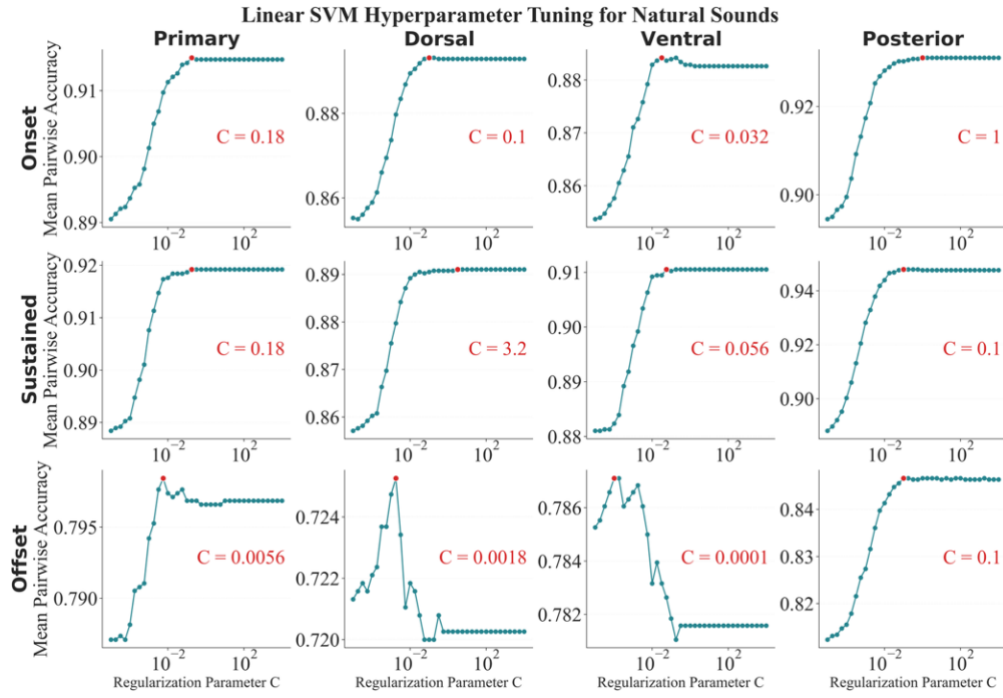
Supplemental Figure S8: Pure Tones Linear SVM Hyperparameter Tuning Curves

Mean pairwise classification accuracy as a function of regularization parameter C ($\log_{10}$ scale) per subregion and spike window. C tuned over 20 logarithmically spaced values from 10^{-5} to 10^4 . Red markers indicate the selected C value (the grid point achieving highest mean accuracy).



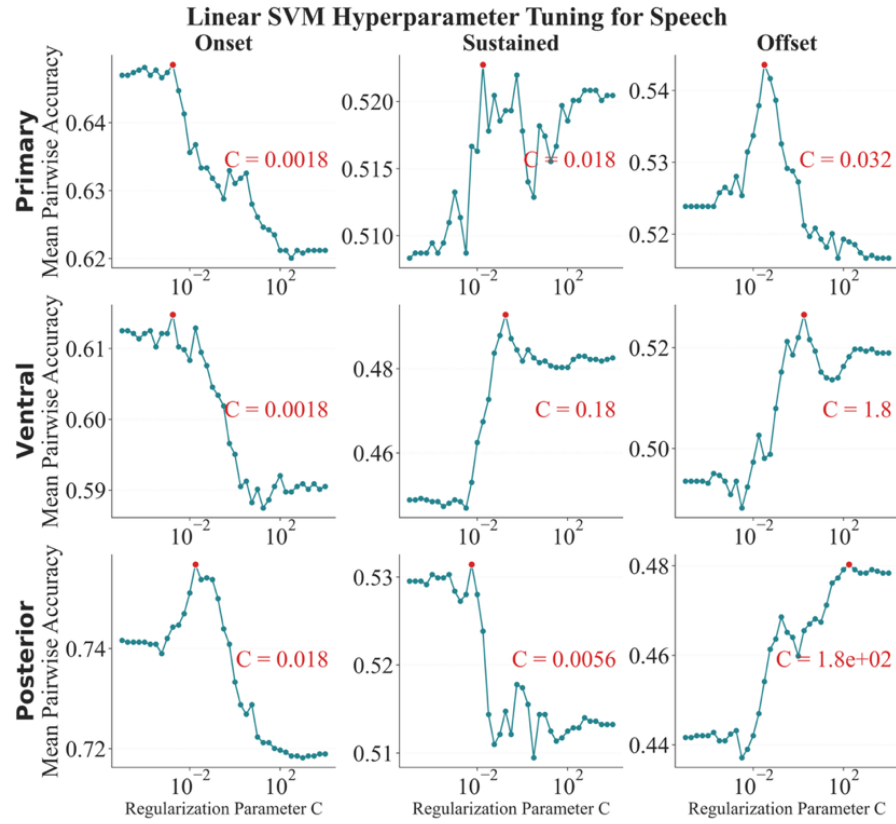
Supplemental Figure S9: AM Linear SVM Hyperparameter Tuning Curves

Mean pairwise classification accuracy as a function of regularization parameter C ($\log_{10}$ scale) per subregion and spike window. C tuned over 20 logarithmically spaced values from 10^{-5} to 10^4 . Red markers indicate the selected C value (the grid point achieving highest mean accuracy).



Supplemental Figure S10: Natural Sounds Linear SVM Hyperparameter Tuning Curves

Mean pairwise classification accuracy as a function of regularization parameter C ($\log_{10}$ scale) per subregion and spike window. C tuned over 20 logarithmically spaced values from 10^{-5} to 10^4 . Red markers indicate the selected C value (the grid point achieving highest mean accuracy).



Supplemental Figure S11: Speech Syllables Linear SVM Hyperparameter Tuning Curves

Mean pairwise classification accuracy as a function of regularization parameter C (log₁₀ scale) per subregion and spike window. C tuned over 20 logarithmically spaced values from 10^{-5} to 10^4 . Red markers indicate the selected C value (the grid point achieving highest mean accuracy).

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